

Real Scene Capture



iPhone
(Real Scene)

3D Scene Reconstruction

VGGT

Point
cloud and
cam info

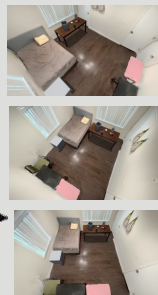
3DGS

Watertight
Mesh

Scene Integration

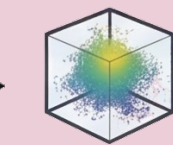
Scene
3DGS

Transformed
3DGS



composite
3DGS

Gaussian Simulator Policy training & evaluation



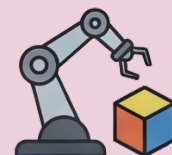
Gaussian Simulator

Vision Data
Sequence



VLA policy
training
& Evaluation

State
Synch



MuJoCo
Physics
Engine

Physics/
Kinematic

Vision Data
Sequence

Robot Real Deployment



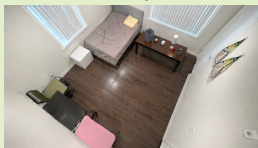
Humanoid
G1

2D Image Editing

Frame
Selection



Nano Banana 2
(Image+Text-to-Image)



Object Reconstruction & Pose Estimation

Canonical object
3DGS & Mesh

Transformed
3DGS

SE(3) Pose
Estimation

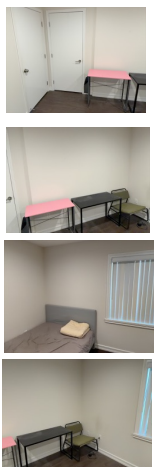
to calculate
layout

Transformed
Object mesh

Physics Merge

VLM physics
parameters
estimation

Collision mesh
With convex
Decomposition
(CoACD)



Dense views